Autodesk® Scaleform®

Scaleform 3Di

Scaleform 3.2 3D .

: Mustafa Thamer

: 2.03

: 2012 5 9

Autodesk® GAMEWARE

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Autodesk Scaleform

Scaleform 3Di

Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

(301) 446-3199

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1 ‹ >

Scaleform® 3Di™ 3D Flash UI .

Scaleform 3.2 Scaleform 3D ActionScript . Scaleform 3.2 ActionScript 2.0 , AS 3.0 3D AS .

(_z, _zscale, _xrotation, _yrotation, _matrix3d, _perspfov) , , HUD UI 3D . 3D movieclip, button textfield .

3D Scaleform Direct Access API C++ . API 3D GFx::Value movieclip, button, textfield , .

, 3D 3D Flash .

2 Flash 10 Ò• g É ú

Scaleform 3Di Flash 10 3D

. 3Di Scaleform AS2 .

2.1

Flash 10 3D 3Di .

1. .

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2. (Back-face culling) .

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2.2

Flash 10 3D

, Scaleform 3Di

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2D 3D , 3D 2D .

3D

. Flash 10 .

3 3D û í Flash ´ ,

Scaleform 3.2 3D Flash 2

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3.1

, Flash 3D

. Scaleform 3.2 3D Flash

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3Di . Flash 3D

, 3D , 3D

. Z( ) , / (UI

) .

,

. , Flash (

Scaleform) . , Flash

. movie clip .
'SWF to Texture' . Scaleform SDK Browser
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3.2 3Di

Scaleform 3.2 , ActionScript Direct Access
API(C++) 3Di . 3Di
. , . , movie clip
. movie clip movie
. , Scaleform
, Scaleform .

, 3Di . , Scaleform UI
(back-face culling) . , UI
. , UI
, , , 3D
UI . , UI
, .

4 3Di API

4.1 ActionScript 2 API

Flash 3Di 2D 3D . ,
3D .
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2 3D . , 3D

(x y , z , 3d), 3D . ,

3D . 3d

null 2D (ActionScript foo._matrix3d = null). z

0 3D .

, Z 2D , 3D 3 x, y, z

. , 3D _zscale ActionScript

.

Scaleform 3D Flash 10 .

, +X , +Y +Z (

) . 55 FOV +Z .

3D Flash .

- Movie clips
- Text fields
- Buttons

3Di ActionScript .

- _z - Z ( ) . 0 .
- _zscale - Z .
  0 .
- _xrotation - X( ) . 0 .
- _yrotation - Y( ) . 0 .
- _matrix3d - 16 float (4 x 4) 3D
  . NULL 3D 2D .
- _perspfov – (Field Of View) . 0
  180 -1 .
  FOV (55) .

gfxExtensions ActionScript true

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movie clip . movie clip

. _xrotation

_yrotation movie clip 3D

. . 3Di

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1

Y movie clip 45 .

```
_global.gfxExtensions = true;
mc1._yrotation = 45;
```

2

Z X .

```
_global.gfxExtensions = true;
sq1._zscale = 500;

function rotateAboutXAxis(mc:MovieClip):Void
{
	mc._xrotation = mc._xrotation + 1;
	if (mc._xrotation > 360)
	   {
	     mc._xrotation = 0;
	}
}

onEnterFrame = function()
{
	rotateAboutXAxis(sq1);
}
```

3

3D 3D .

```
function PixelsToTwips(iPixels):Number
{
        return iPixels*20;
}


function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [        1, 0, 0, 0,
                    0, 1, 0, 0,
                    0, 0, 1, 0,
                    tX,tY,tZ,1];
    return matrixC;
}

this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);
```

4.2 C++ 3Di

4.2.1 Matrix 4x4

4x4 Matrix4x4 .
3D (,)
. Render_Matrix4x4.h .

4.2.2 Direct Access

GFx::Value Direct Access 3Di . 3D
API . 3Di ActionScript
.

GFx_Player.h GFx::Value::DisplayInfo .

```
void   SetZ(      Double   z)
void   SetXRotation(         Double   degrees)
void   SetYRotation(         Double   degrees)
void   SetZScale(        Double   zscale)
void   SetFOV(      Double   fov)
void   SetProjectionMatrix3D(const                  Matrix4F *pmat)
void   SetViewMatrix3D(const              Matrix3F *pmat)
bool        SetMatrix3D(    const   Render::Matrix3F      & mat)

Double  GetZ()       const
Double  GetXRotation()         const
Double  GetYRotation()         const
Double  GetZScale()        const
Double  GetFOV()       const
const    Matrix4F* GetProjectionMatrix3D() const
const    Matrix3F* GetViewMatrix3D() const
bool      GetMatrix3D(Render::Matrix3F            * pmat) const
```

4.2.3 Render::TreeNode

3D Render::TreeNode

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GFx_Player.h

```
void SetProjectionMatrix3D(const Matrix4F& m)
void SetViewMatrix3D(const Matrix3F& m);
```

4.2.4 : Direct Access API n É E R a Û 3 ¾

```
Ptr<GFx::Movie> pMovie = …;
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
        GFx::Value::DisplayInfo dinfo;
        bOK = tmpVal.GetDisplayInfo(&dinfo);
        if (bOK)
        {  // set perspectiveFOV to 150 degrees
              Double oldPersp = dinfo.GetFOV();
              dinfo.SetFOV(150);
              tmpVal.SetDisplayInfo(dinfo);
        }
}
```

5 R

Direct Access API FOV( )

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FOV 0 FOV .

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. z

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. 0 180 . 1

, 179 .

z 2: ` ´ u Æ r R Flash „ ò

z 3: -80 • x ² `´ Ò ? R ³ E(55 c)

FOV (5) .

. ActionScript _perspfov

-1 . C++ Direct Access API

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// 1. FOV -1

```
GFx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
  GFx::Value::DisplayInfo dinfo;
  bOK = tmpVal.GetDisplayInfo(&dinfo);
  if (bOK)
  {
    dinfo.SetPerspFOV(-1);
    tmpVal.SetDisplayInfo(dinfo);
  }
}
```

// 2.

```
void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)
{
  const float nearZ = 1;
  const float farZ = 100000;
```

```
   float DisplayHeight = fabsf(visFrameRectInTwips.Height());
   float DisplayWidth = fabsf(visFrameRectInTwips.Width());
   matPersp-      >OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
GFx::Value tmpVal;
bool bOK = pMovie-      >GetVariable(&tmpVal, "root.mc");
if (bOK)
{
   GFx::Value::DisplayInfo dinfo;
   bOK = tmpVal.GetDisplayInfo(&dinfo);
   if (bOK)
   {
      Matrix4F perspMat;
      MakeOrthogProj(PixelsToTwips (pMovie-                    >GetVisibleFrameRect()), &perspMat);
      dinfo.SetProjectionMatrix3D(&perspMat);
      tmpVal.SetDisplayInfo(dinfo);
  }
}
```

.

```
Ptr<Movie> pMovie = …
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie-                    >GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()),   fabs(visFrameRectInTwips.Height()),
nearZ, farZ);
pMovie-  >SetProjectionMatrix3D(ortho);
```

5.1 AS3

ActionScript 3 PerspectiveProjection

.

Example:

```
import     flash.display.Sprite;

var    par:Sprite        =  new  Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0,             0,    100,    100);
par.x     =  100;
```

```
par.y     =  100;
par.rotationX         =  20;

addChild(par);

var    chRed:Sprite       =  new  Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0,            0,    100,    100);
chRed.x    =  50;
chRed.y    =  50;
chRed.z    =  0;

par.addChild(chRed);

var    pp3:PerspectiveProjection=new             PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection                    =  pp3;
```

6 X e # X 3D

Scaleform 3Di 3D UI
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Scaleform 3Di NVIDIA PC 3D Vision Kit 3D
. NVIDIA
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. Display
, . Scaleform
4.0 PS3 TinyPlayerStereo
. TinyPlayerStereo Scaleform .
Apps\Samples\GFxPlayerTiny\GFxPlayerTinyPS3GcmStereo.cpp

6.1 NVIDIA 3D Vision

6.1.1 3D Vision ó

NVIDIA 3D Vision Kit 3Di
. NVIDIA . 3D '
3D ' . NVIDIA Stereoscopic 3D Test
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6.1.2 e *

, D3D9 Scaleform Player .
Automatic 3D Vision Direct3D Scaleform Player
. Scaleform Player -f
. NVIDIA 3D
. 3D (CTRL + T)
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6.1.3 ñ ´ X 8

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\+ -
. 3D Vision ,
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3D Vision IR (CTRL+F3)/(CTRL+F4)
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Scaleform Player .
(CTRL+F5)) (CTRL+F6) , .
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NVidia "advanced in-game settings
" . , "*Stereoscopic 3D*" "*Set up stereoscopic 3D*"
"*Set Keyboard Shortcuts*" . (CTRL+F5) (CTRL+F6)
. (CTRL+F7) .
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(CTRL+F5) .

10 15 .

(CTRL+F5) (CTRL+F6)

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0 . , ,

(CTRL+F7) .

NVIDIA 3D API 3D

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API

http://developer.nvidia.com/object/nvapi.html.

Designing for stereo( )

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform API

Scaleform 4.1 Render::HAL 3D API

. 3D

HDMI 1.4 .

6.2.1 , Scaleform

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Scaleform 4

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Render::Viewport::SetStereoViewport .

SF4 Viewport Stereo .

```
// (
.
View_Stereo_SplitV          = 0x40,
View_Stereo_SplitH          = 0x80,
View_Stereo_AnySplit        = 0xc0,

void SetStereoViewport(unsigned display);
```

6.2.1 Scaleform Stereo œ U

```
Render::StereoParams s3DInfo;
s3DInfo.DisplayDiagInches = 46;              // 46 TV
s3DInfo.DisplayAspectRatio = 16.f / 9.f;
s3DInfo.Distortion = .75;                    // 0 1
s3DInfo.EyeSeparationCm = 6.4;               // 6.4cm .
pRenderHAL- >SetStereoParams(s3DInfo);
```

Display ,
Display . Scaleform
. :

```
pMovie- >Advance(delta);

pRenderer- >GetHAL()- >SetStereoDisplay(Render::StereoLeft);
pMovie- >Display();

pRenderer- >GetHAL()- >SetStereoDisplay(Render::StereoRight);
pMovie- >Display();
```

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Scaleform SDK 3D Flash (SWF, FLA) 3Di

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. *C:\Program Files (x86)\Scaleform\GFx SDK 4.2\Bin\Data\AS2 or AS3\Samples*

z 4: 3DGenerator

z 5: 3D É z F

z 6: 3D a c F